



Liquid fuels demand and supply model

Week 1 analytical pack

South Africa | Petrol and diesel

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com

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Size the fuel market, reconcile supply, then test Vopak access

Situation → Complication → Resolution | evidence, actions and delivery roadmap

SCR reference	Evidence and required next step	Proposed lead	Timing and output	Deep dives
S1 Situation	Demand baseline and regional market 2022 provinces are observed; 2023-2024 are estimated separately. Resolve source versions before forecasting.	Manish Nigel review	W1-W2 Sourced demand baseline	4 / 5
S2 Situation	Domestic supply and finished imports SARS 2024 net diesel imports: 9.997 bn L. Match actual production, stocks and sales coverage; port allocation remains open.	Manish Nigel access	W1-W3 Fuel balance + port bridge	3 / 6
C1 Complication	Demand drivers Monthly activity to Jul 2026; GDP Q2 2026; inland prices Oct 2026. Define activity-to-litres bridges before calibrating cases.	Manish Nigel review	W1 define; W2-W4 test Lever-to-demand bridge	7
C2 Complication	Refinery availability Capacity history and reported operator output are distinct. Align fiscal/calendar years and product yields; redevelopment stays conditional.	Manish Henry / Nigel	W1-W3 Plant output + imports	8
C3 Complication	Competing supply routes Compare Durban, Matola and Walvis Bay for one customer. Test full route cost, capacity and customer access.	Manish Nigel / Henry	W1 scope; W2-W4 test Delivered R/litre + access	9 / 10
R1 Resolution	Vopak share and additional demand Match unique deliveries to regional demand. Test additional volumes against physical and commercial access.	Nigel client data Manish analysis	W1 request; W2-W4 test Served + contestable litres	11 / 12 / 13
R2 Resolution	Storage and optional investment Use published tanks as context. Assess additional services only after the incremental customer-flow case.	Nigel priority Manish / Henry	W4-W6 if prioritised Flow-to-stock / investment	14 / 15

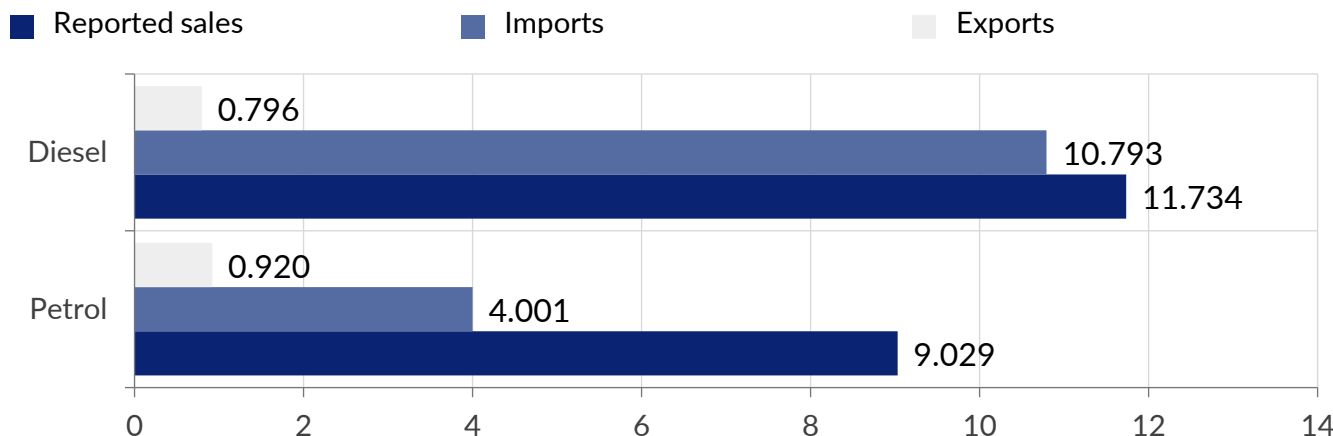
Petrol + diesel; jet separate. Owners and weeks are proposed. Open gaps and illustrations remain explicitly labelled.

Customs records 10.0 billion litres of net diesel imports in 2024

National petrol/diesel | 2024, billion litres

Production + imports – exports – stock build = consumption

Compare consumption with reported sales; explain coverage and the residual.



Net imports: petrol 3.081, diesel 9.997; combined 13.078 bn litres.

Primary trade | SARS customs

Diesel imports 10.793 bn L.
Government rounds to 10.800; FIASA 14.793.
Use SARS; retain the source discrepancy.

2024 accounting remains open

Sales less net imports = 7.685 bn L.
Balancing requirement, not production.
Production, stocks and coverage unresolved.

What the evidence shows

01 | Customs supports the trade selection

SARS records diesel imports of 10.793 and exports of 0.796 bn litres in 2024. Net imports are 9.997; use customs as the primary trade source.

02 | Reported sales remain a proxy

FIASA reports petrol/diesel sales of 20.763 bn litres for 2024. Editions differ; this demand proxy remains unverified and is not imports plus exports.

03 | Production and stocks remain open

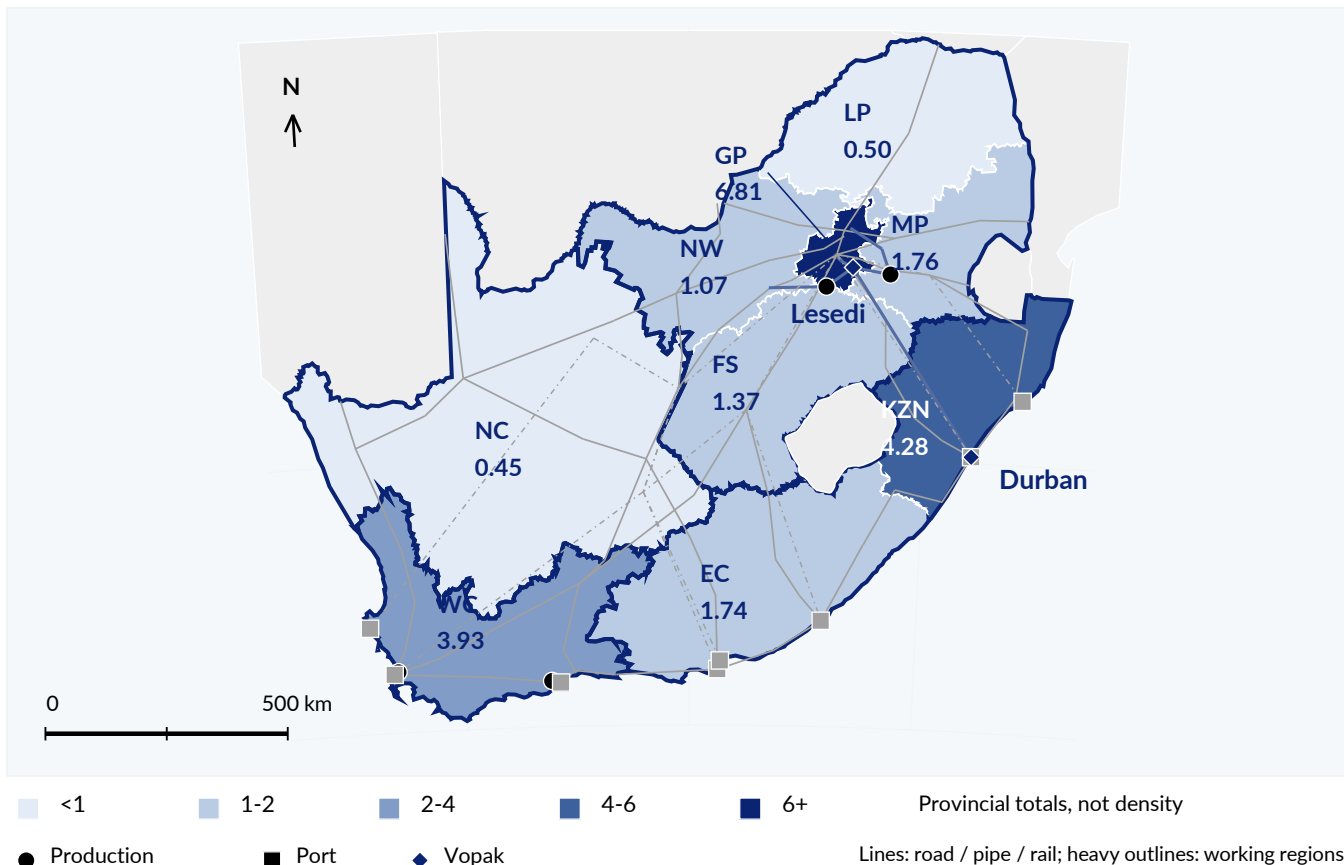
Production and stocks remain unresolved. The proposed 3-4 bn litre diesel gap is a hypothesis, not measured consumption.

To resolve next

Manish: rebuild the balance CSV from SARS; match product, year, actual output and stock changes. Nigel: agree the sales selection and obtain customer/port evidence.

Three provinces account for 69% of reported fuel sales

Petrol + diesel sales by province | 2022, bn litres/year



Sales totals, not density. Vopak locations highlighted; detailed operator inventory on pages 12 and 14. Jet excluded.

Source: DMPR 2022 sales; geoBoundaries/OCHA/MDB 2020 provinces (CC BY 3.0 IGO); operator sources on slide 12.

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What the evidence shows

01 | Demand is concentrated

Gauteng, KwaZulu-Natal and Western Cape account for 68.6% of reported 2022 petrol/diesel sales.

02 | Four reporting regions

Eastern coast: EC/KZN. Inland: GP/FS/LP/MP/NW. Western coast: WC. Other: NC. Boundaries are reporting groups.

03 | Sales locate the historical market

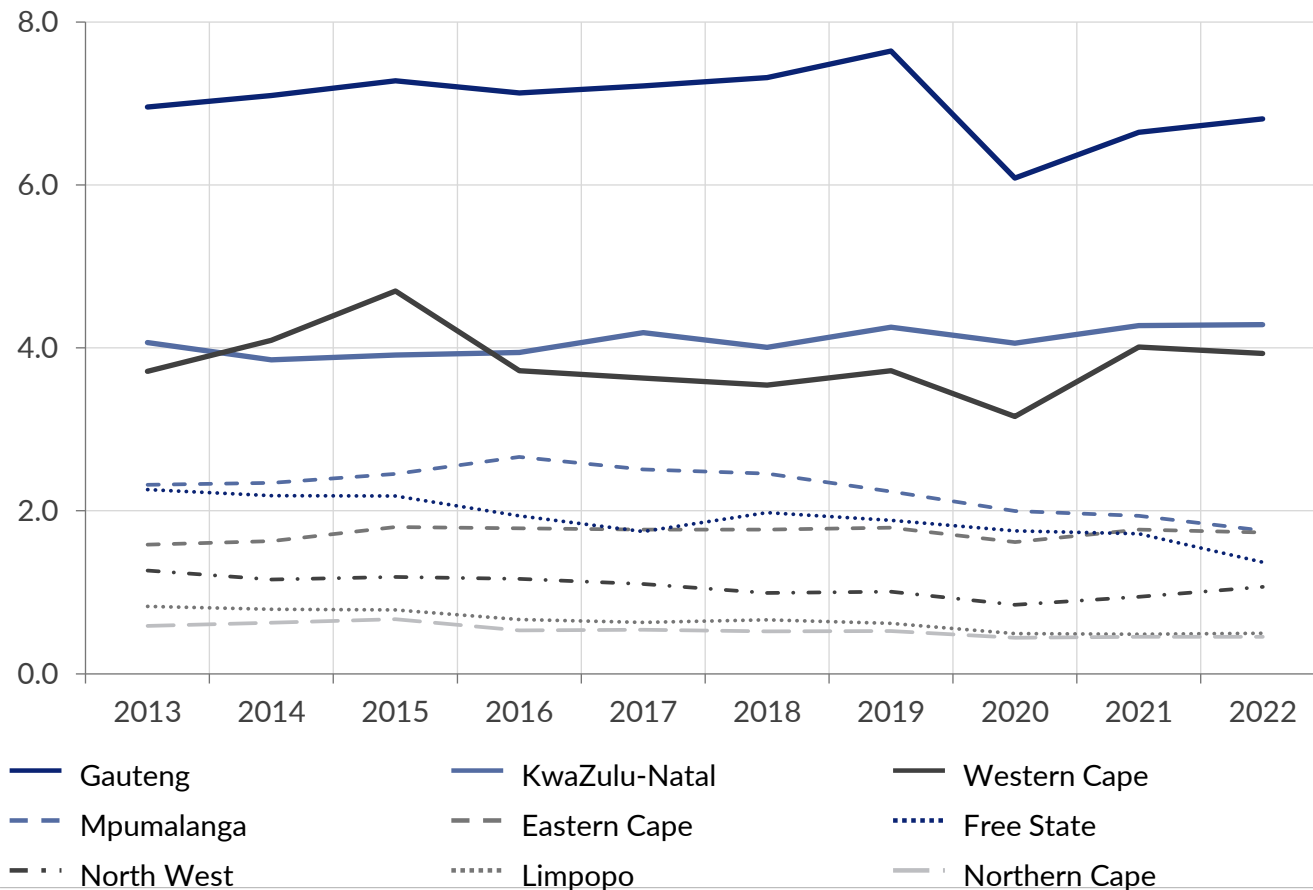
Colours show provincial sales totals, not density or customer catchments. Only Durban and Lesedi are labelled; competitor detail is on pages 12 and 14.

To resolve next

Nigel: confirm customer markets for Durban and Lesedi. Manish: retain the observed 2022 map; present later estimates separately with national-source flags.

Provincial sales show different demand patterns over time

Annual provincial sales | 2013–2022, bn litres/year



What the evidence shows

01 | Gauteng remains the largest market

The reported series places Gauteng above the other provinces throughout 2013–2022. Volumes differ across the smaller markets.

02 | Later provinces are estimates

2024 petrol estimates: Gauteng 3.47, KZN 1.47, Western Cape 1.41 bn L. These hold 2022 shares; no later observed provincial sales were found.

03 | Historical differences are traced

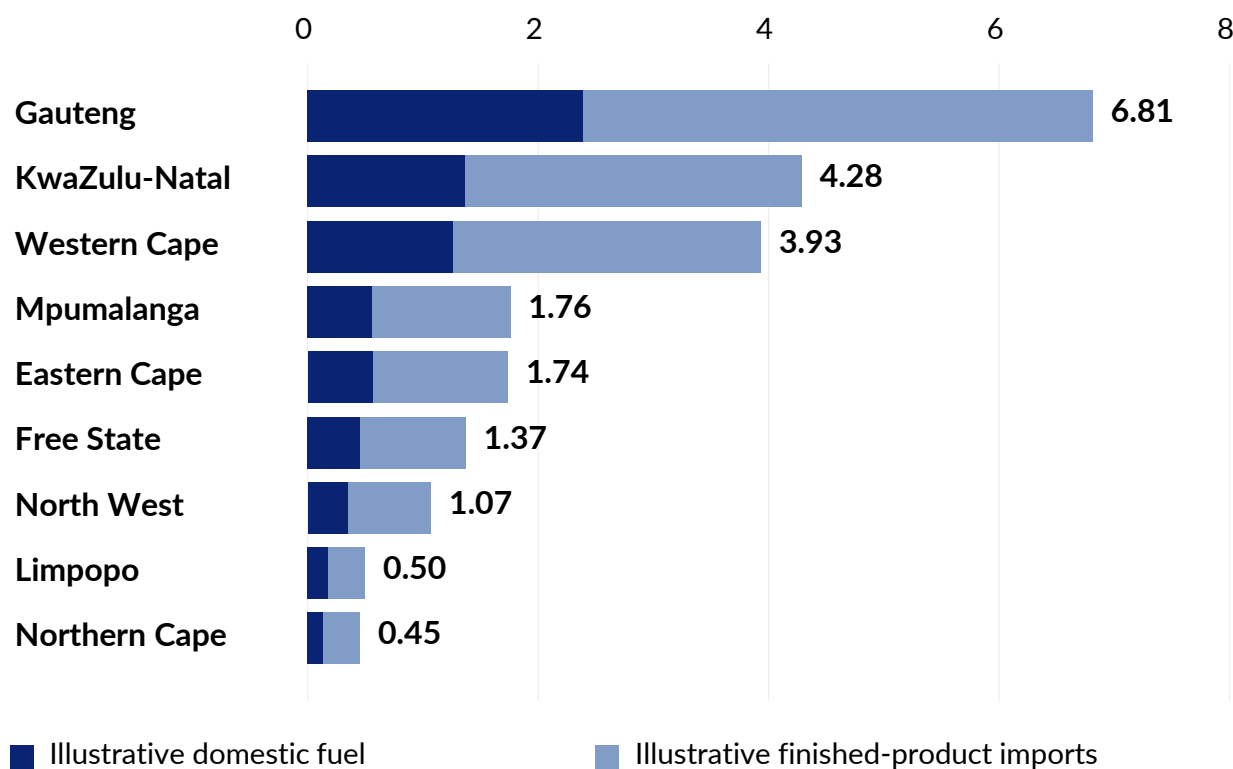
The 2013 national parser is corrected on the analyst branch. 2014/2015 source versions differ; 2018 district coverage is incomplete. Preserve the flags.

To resolve next

Manish: reconcile source selections and retain estimate labels. Nigel: decide whether estimated petrol volumes belong in the outlook; diesel booking shares are less stable.

Provincial demand is known; its supply origin still needs evidence

2022 sales totals | ILLUSTRATIVE domestic/import allocation



Units: bn litres/year. Imported/domestic segments are a scenario illustration, not measured provincial flows.

What the evidence shows

01 | Demand totals are sourced

The provincial bars use reported 2022 petrol/diesel sales.

02 | Origin shares are illustrative

Domestic/import fractions are authored examples; no actual provincial origin allocation is established.

03 | Imports need a national balance

Exports and stock movements must be reconciled before a sourced import requirement can be presented.

To resolve next

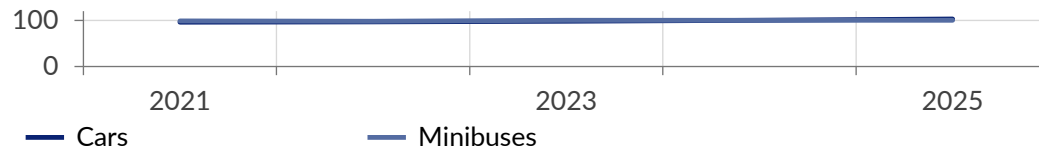
Manish: populate the matched-year national balance first; retain provincial origin as unknown until supported.

Observed activity helps define the fuel levers; intensities remain open

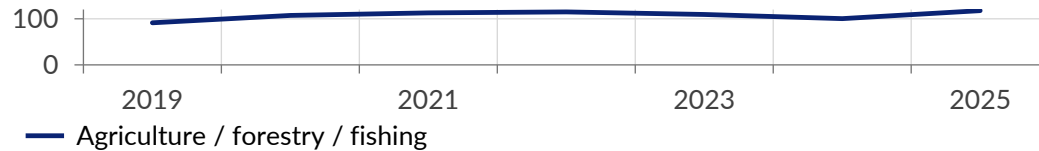
Potential demand levers | 2024 = 100; price panel uses nominal R/litre

Observed proxies, not fuel-volume effects. Activity axes 0–120; EV 0–400; prices 0–30 R/L.

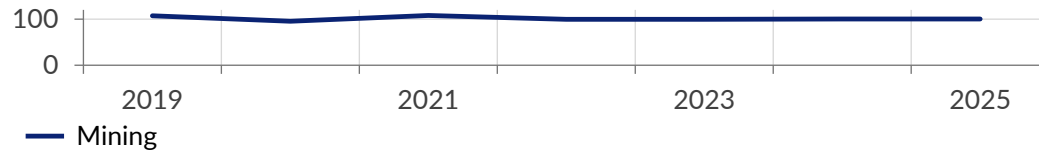
Passenger travel | all-fuel stock, Dec



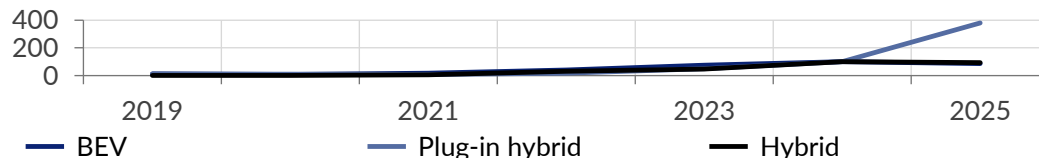
Agriculture | real value added, CY



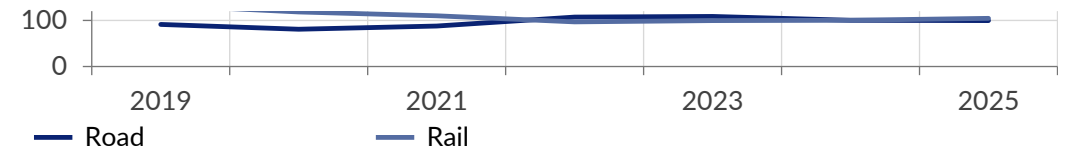
Mining | production volume, CY



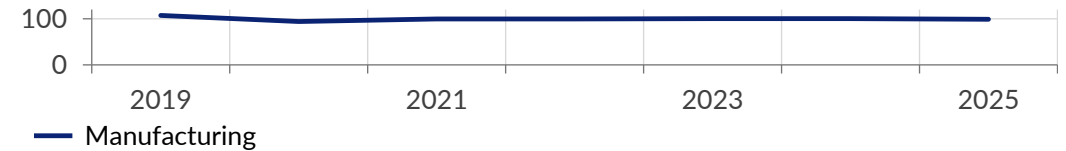
Electrification | annual new sales, CY



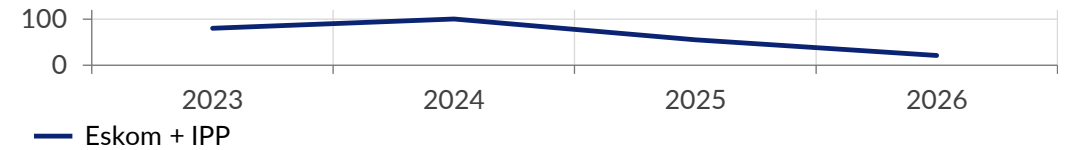
Freight demand | payload, CY



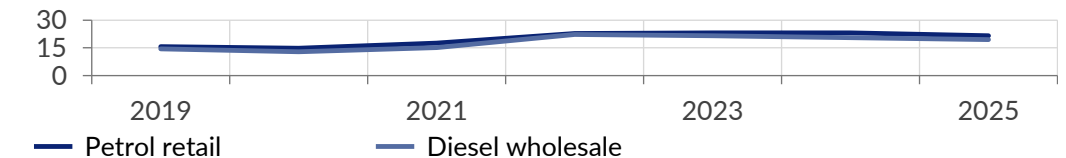
Manufacturing | production volume, CY



Diesel power | OCGT generation, FY



Price response | annual nominal R/litre



Latest staged: activity Jul 2026; GDP Q2 2026; inland prices Oct 2026. Charts use complete years. Fuel intensities remain open; FY ends March.

Source: NaTIS; Stats SA P0441 / P2041 / P3041.2 / P7162; Eskom; naamsa; DMPR / CEF. Snapshot manifest in notes.

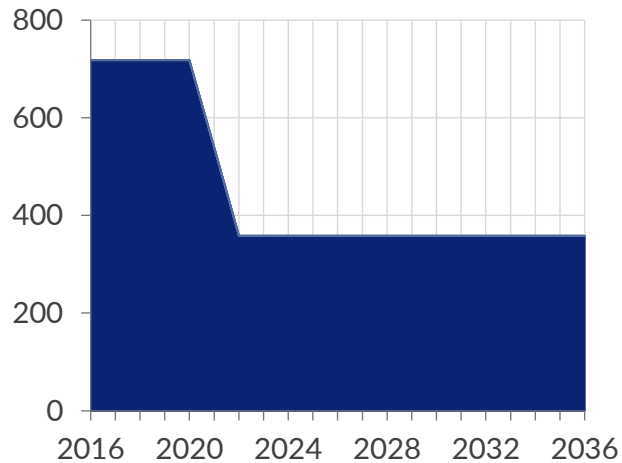
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Refinery capacity has fallen; redevelopment remains conditional

Published footprint + illustrative scenarios | thousand bbl/day

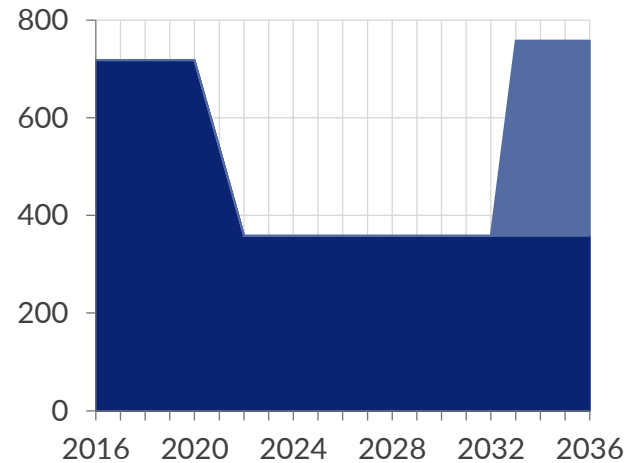
History 2016–2025 is sourced; 2026 onward is illustrative, not a forecast

A | No new refinery capacity



2036: 358 thousand bbl/day

B | Conditional CEF redevelopment



2036: 758 thousand bbl/day

Reported FY2024 output: Secunda 29.1; Natref Sasol share 17.8 million barrels.

All products; year to June, not calendar-year petrol/diesel. B adds 400 from 2033, conditional on FID.

What the evidence shows

01 | Capacity has fallen

Published capacity falls from 718 to 358 thousand bbl/day between 2020 and 2022. Redevelopment remains conditional.

02 | Reported output is different

FY2024 Secunda output is 29.1 million barrels; Natref is 17.8 for Sasol's share. Both cover all products and a year to June.

03 | A product bridge is needed

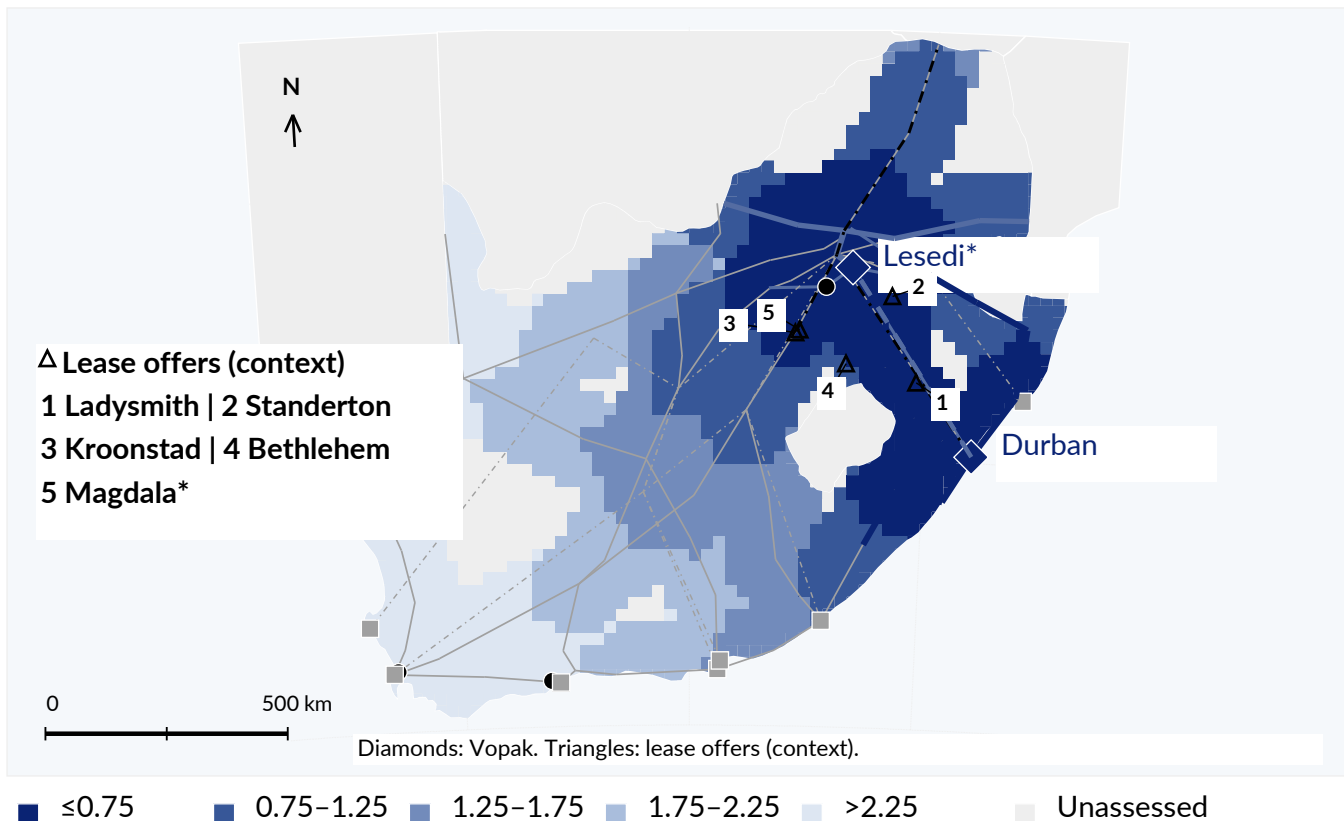
Match product yields, calendar years and downtime before calculating import effects. Capacity does not measure actual output.

To resolve next

Manish: match operator output to product/year and document Natref ownership basis. Henry: review technical mechanisms. Nigel: agree refinery scenario timing.

Transport cost shapes the market Durban and Lesedi can reach

Illustrative road cost, R/litre



Example rate: R0.15/L dispatch + R0.002/L/km. Grey = unassessed.

Illustrative road cost only. Pipeline/rail access and commercial rights are not priced.

Source: Existing appendix routes; illustrative candidate connections, not verified catchments. Natural Earth 1:50m.

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What the evidence shows

01 | Darker means lower transport cost

Cells show example road cost from Durban or Lesedi. The rate is illustrative and excludes fuel price, storage and pipeline tariffs.

02 | Grey means unassessed

Grey areas have no assessed road connection under this illustration. They are not confirmed to be inaccessible.

03 | A cost surface is a first screen

Pipeline and rail lines are context. Route capacity, product compatibility and customer rights still determine serviceable demand.

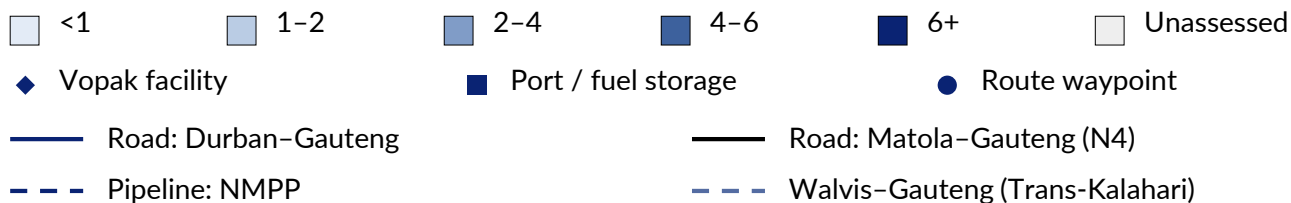
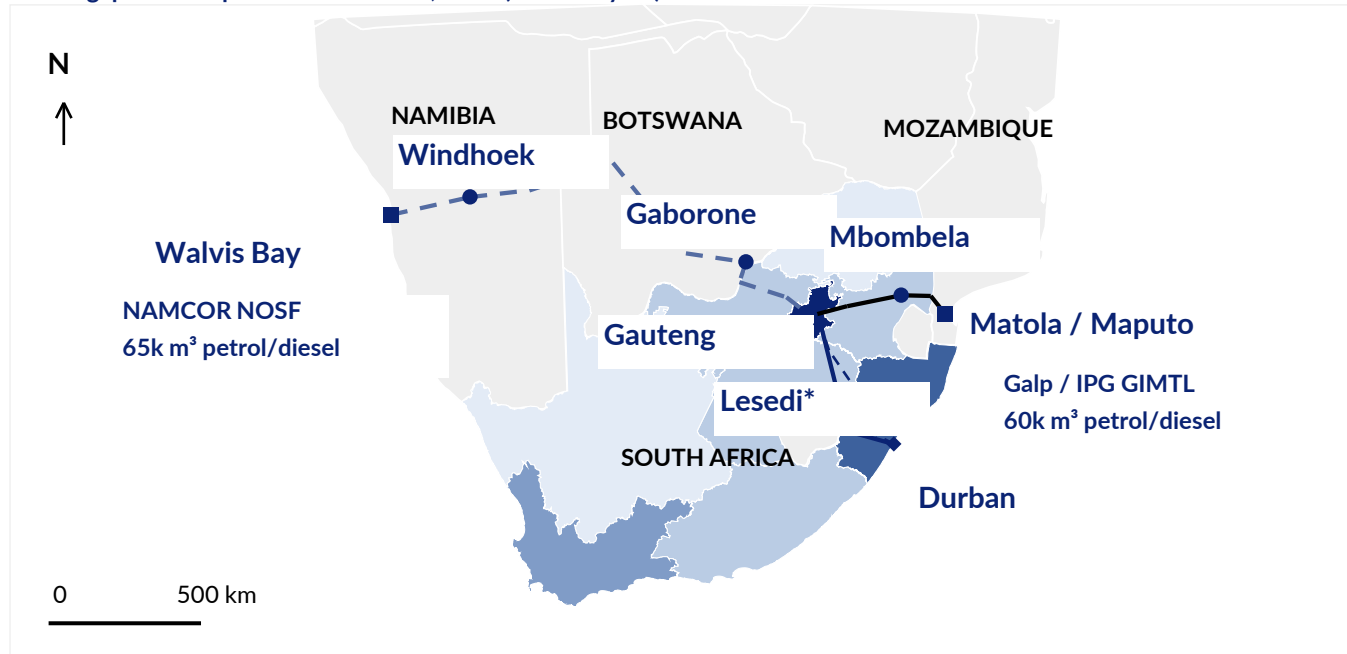
To resolve next

Nigel: select the first customer destination and cost threshold. Manish: replace example rates with dated route costs and access evidence.

Compare Durban, Matola and Walvis Bay for the same inland customer

Durban, Matola and Walvis Bay | schematic corridor comparison

Shading: provincial petrol + diesel sales, 2022 (bn litres/year)



Routes schematic; access unverified. Colour measures provincial totals, not density or Vopak share.

Source: DMPR 2022 sales; NAMCOR NOSF; Galp GIMTL; WBCG / TRAC corridors; Natural Earth. Capacities and URLs in notes.

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What the evidence shows

01 | Vopak has a South African footprint

Durban and Lesedi are highlighted against 2022 provincial sales. Colour shows sales totals; it does not establish Vopak market dominance.

02 | Matola has published fuel storage

Galp/IPG GIMTL lists 60,000 m³ of petrol/diesel tanks, plus 6,000 m³ LPG excluded here. N4 is the eastern inland comparison route.

03 | Walvis has storage and handling

NAMCOR NOSF lists 75,000 m³ in total, including 65,000 m³ petrol/diesel. Trans-Kalahari is the western inland comparator.

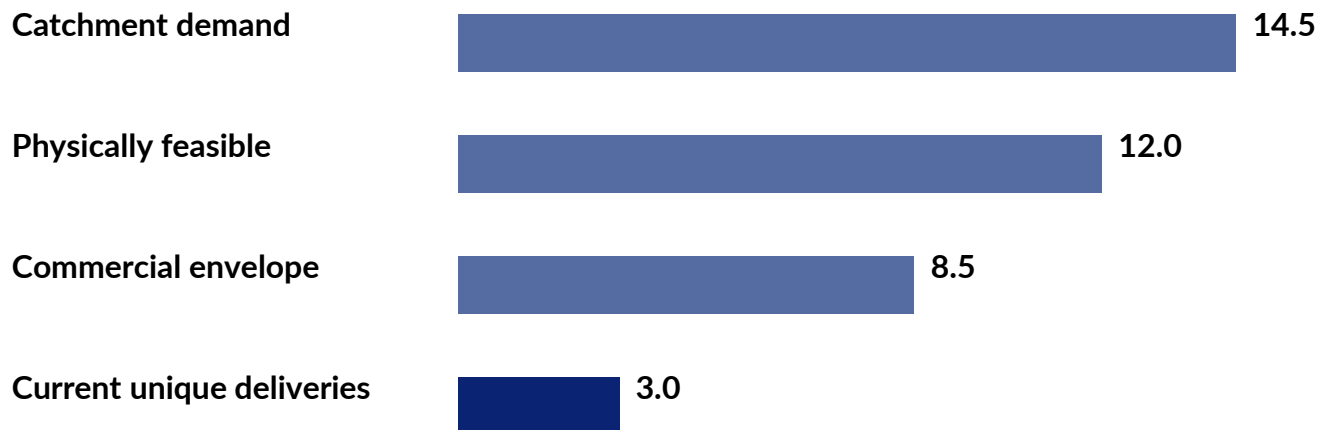
To resolve next

Manish: verify available tanks, customer access and full delivered costs. Source Namibia/Mozambique demand before extending the heatmap; grey means unassessed.

Separate market demand, accessible volume and unique Vopak deliveries

Market screens | illustrative bn litres/year

Successive screens; these volumes are illustrative



Count Durban–Lesedi transfers once

Durban 2.8 + Lesedi 2.0 – shared transfer 1.8 = unique 3.0

Demand ≠ terminal receipts ≠ storage capacity. Incremental opportunity is on page 13.

What the evidence shows

01 | Count customer demand once

Example receipts of 2.8 at Durban and 2.0 at Lesedi include a shared 1.8 transfer. Unique demand served is therefore 3.0 bn litres.

02 | Accessibility precedes opportunity sizing

Example catchment demand 14.5 → physically feasible 12.0 → commercial envelope 8.5. Each screen requires evidence.

03 | Examples are separate from sales

These market volumes illustrate the calculation. They are not the reported 2022 regional totals or actual Vopak deliveries.

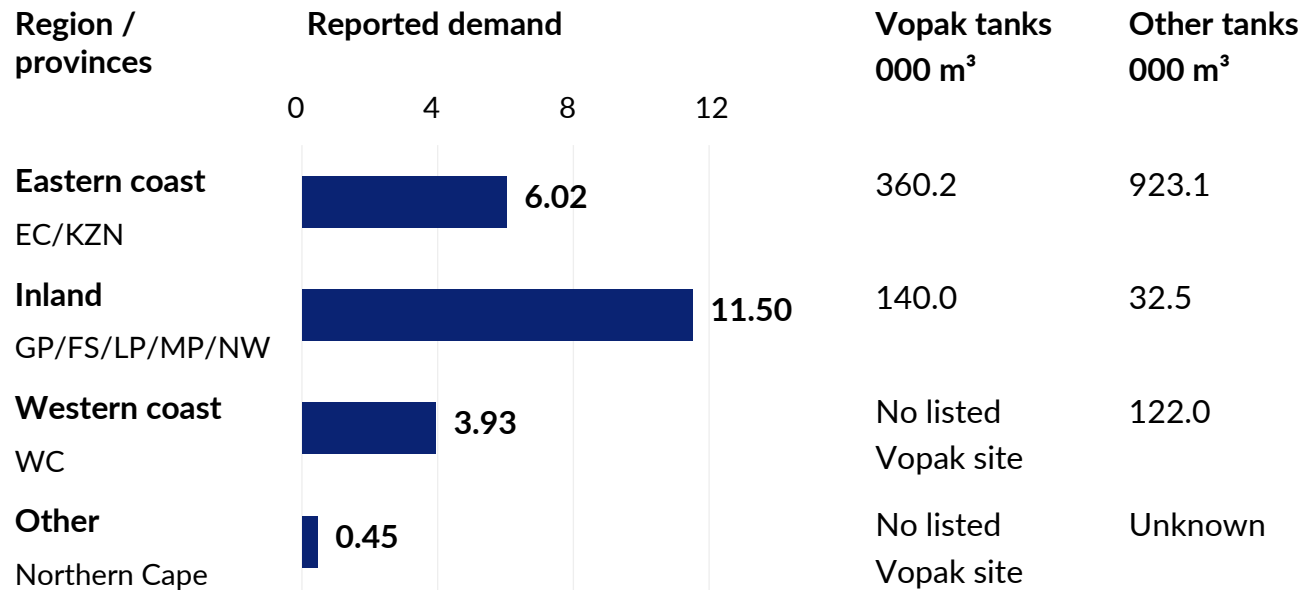
To resolve next

Nigel: request customer deliveries, destinations and transfer records. Manish: calculate unique served demand before estimating additional access.

Compare regional demand with Vopak's published storage footprint

All four regions | reported 2022 demand, bn litres/year

National total 21.90 bn L | petrol + diesel; jet excluded



Published capacity is a provisional footprint measure, not annual demand served.

Partial gross inventory; mixed products and unknown Sasol/Transnet capacities.

Fuel-compatible capacity share and actual Vopak demand share remain unestablished.

[1] Vopak [2] Bidvest [3] Burgan [4] Shell [5] NERSA [6] Sasol [7] Transnet

Source: DMPR 2022 provincial petrol/diesel sales; public operator sources [1-7], checked 6 Oct 2026. Share not established.

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Regional competitor evidence

01 Eastern coast

Vopak: Durban [1]. Bidvest: Durban and Richards Bay [2]; mixed-product capacity.

Transnet: Ladysmith lease tanks [7].

02 Inland

Vopak: Lesedi [1]. Bidvest: Isando [2].

Sasol: Alrode, Pretoria West, Waltloo, Sasolburg [6].
Transnet: Tarlton / Jameson Park + four lease sites [7].

03 Western coast

Burgan Cape: Cape Town petrol/diesel terminal [3].
Confirm current tank availability and customer access.

04 Other / Northern Cape

Operator coverage incomplete. Check Shell site notices [4] and NERSA licences/access records [5].

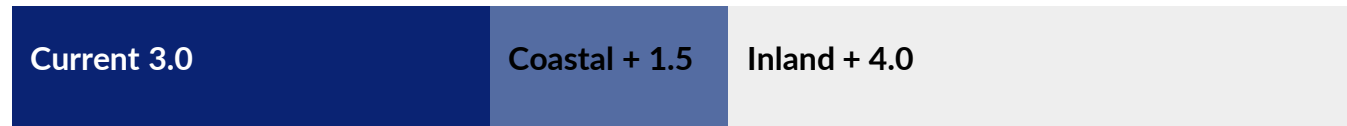
The illustrative additional opportunity is 5.5 billion litres a year

Accessibility gates and volume milestones | illustrative bn L/year

ILLUSTRATIVE | each increment requires evidence; not cumulative terminal receipts

5.5 bn L/year additional candidate

3.0 current example + 5.5 additional = 8.5 total envelope



What must hold before any additional litres are counted?

Competitive cost

Usable capacity

Customer access

Unique deliveries

Detailed customer-by-route gates belong in the evidence register.

Illustration only; 5.5 is neither a forecast nor a confirmed capturable market.

What the evidence shows

01 | Reach is necessary, not sufficient

Accessibility map p9 and competing routes p10 locate candidate access. Neither currently links a verified cost threshold to customer litres.

02 | Example milestones are conditional

The current example is 3.0. Coastal candidate 1.5 raises it to 4.5; inland candidate 4.0 raises it to 8.5. These are authored volumes.

03 | Every increment needs four gates

Evidence cost, route/tank capacity, customer rights and unique deliveries for the same product and period. Unassessed litres remain explicit.

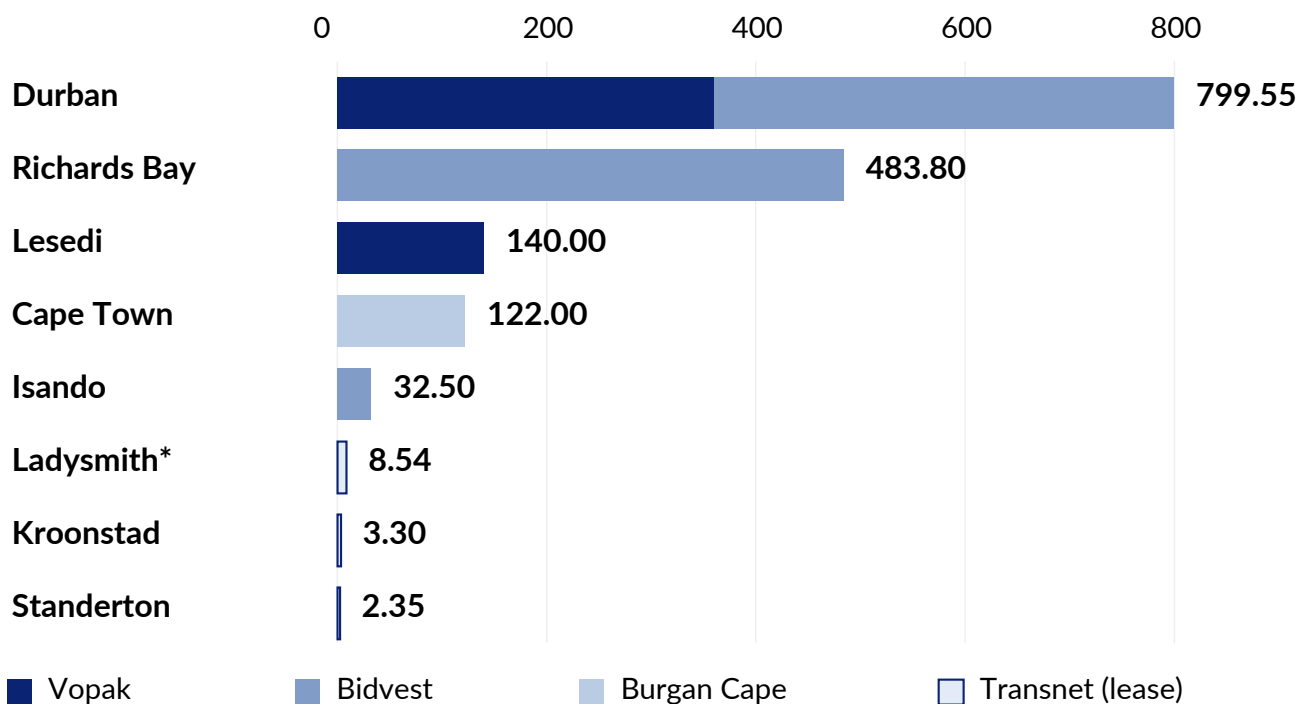
To resolve next

Nigel: agree destinations and acceptable delivered cost.
Manish: build a customer-by-route gate table; record passed, failed and unassessed litres, with sources and conditions.

Durban leads the published storage subset

Published storage by location | thousand m³; mixed capacity bases

One shared scale for every location



Small lease bars reflect smaller tank volumes; they are not operating supply.

*Ladysmith excludes 0.474 intermixture. Unquantified: Bethlehem, Magdala, Sasol, Tarlton, Jameson Park.

Source: Vopak; Bidvest; VTTI Borgan Cape; Transnet lease RFPs (5 Oct 2026), pp. 2-3. Checked 6 Oct 2026.

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What the evidence shows

01 | Capacity is a stock

Published tank volumes are not annual fuel deliveries or market share.

02 | The inventory is partial

Missing capacities are unknown, rather than zero; mixed product scopes limit comparison.

03 | Leases are conditional

Transnet lease tanks are not counted as currently available operating supply.

To resolve next

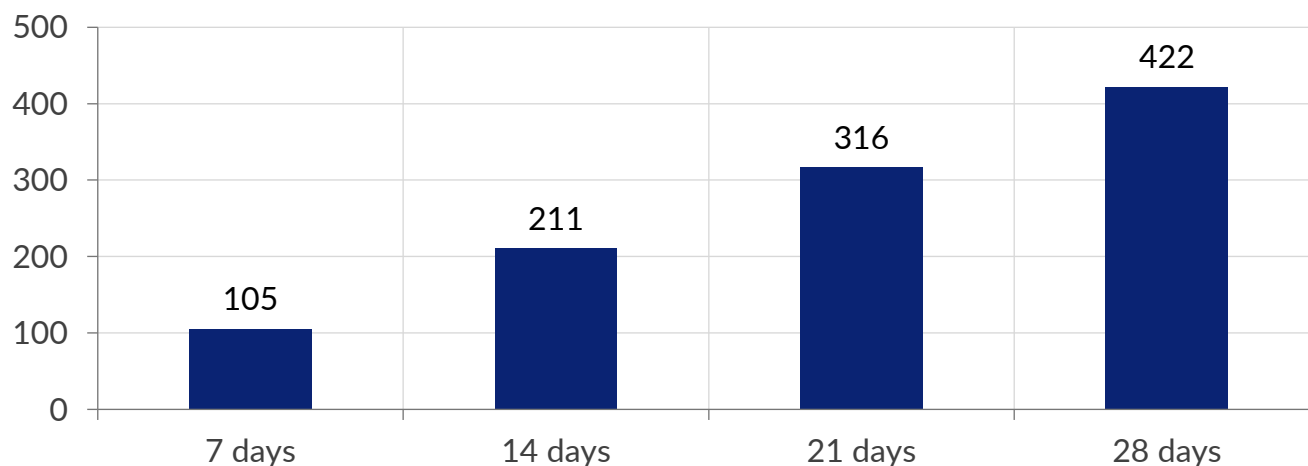
Manish: verify product compatibility, capacity basis and operating status. Nigel: confirm usable client capacity.

Translate additional fuel flows into a preliminary inventory requirement

Working inventory sensitivity | illustrative, thousand m³

ILLUSTRATIVE | additional 5.5 bn litres/year; uniform daily flow

More inventory days require more working stock



Working m³ = annual litres ÷ 365 ÷ 1,000 × inventory days

At 14 days: approximately 211 thousand m³ of working inventory.

New tanks depend on usable spare capacity, peaks, product segregation and dispatch limits.

What the evidence shows

01 | Inventory days set working stock

At the illustrative 5.5 bn litres/year, 14 days needs about 211 thousand m³. More inventory days require more stock.

02 | Existing tanks may cover part

Usable spare tanks may absorb this inventory. Published gross capacity does not establish spare petrol/diesel space.

03 | Sizing needs an operating case

Test peak receipts, product segregation, tank heels and dispatch limits. This is a sensitivity, not an investment recommendation.

To resolve next

Nigel: confirm candidate flows and inventory policy.
Manish and Henry: test usable spare tanks, peak flows and operating allowances before sizing new capacity.

Agree priorities and next steps

Confirm demand, routes and access

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com



Presentation palette | colours and HEX codes

Reference | colours used in this presentation

Core presentation colours

	Vopak blue #0A2373		Secondary blue #546CA2		Text / ink #000000		White #FFFFFF
	Unassessed / neutral #EEEEEE		Header divider #A0A0A0		Body divider #B8B8B8		Frame divider #B0B0B0

Sales-map colours | provincial totals, bn litres/year

	Below 1 #E3EBF6		1 to below 2 #BACCE4		2 to below 4 #809CC7		4 to below 6 #3D619D
	6 and above #0A2373		Map background #F5F8FB				

Other template theme colours

	Theme dark 2 #11151A		Theme light 2 #E7E6E6		Accent 3 #BDBEC1		Accent 4 #404040
	Accent 5 #767676		Accent 6 #D9D9D9				

Typeface: Lato. Sales colours show totals, not density or market share. Grey denotes unassessed demand.

Source: supplied Vopak template; pptx/brand_configs/vopak.py; provincial_demand_map.py.

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